

626 Exchange Seats

PLATFORM

Platform: LeetCode

DIFFICULTY

MEDIUM

DATE

Solved: June 29, 2026

Input

- 1 table — Seat with columns id, student
- id is the primary key, starts from 1 and increments continuously

Output

- Swap student names between every two consecutive seats
- If total seats is **odd**, last seat remains unchanged
- Result ordered by id ascending

Key Insights

- This is a **computed column** problem — we change the id value, not the student, to achieve the swap effect (clever trick!)
- Odd ids move forward (id+1), even ids move backward (id-1)
- Special edge case: last seat when count is odd → must check against MAX(id)

Observations

- Swapping id values (instead of student values) and then ORDER BY id automatically rearranges students correctly
- The last odd seat (when total is odd) has no pair → must remain unchanged
- $id \% 2 = 1 \rightarrow \text{odd}$; $id \% 2 = 0 \rightarrow \text{even}$

Approach

3 conditions checked in priority order:

- Odd id AND not the last seat → new id = id + 1
- Even id → new id = id - 1
- Otherwise (last odd seat) → new id = id (unchanged)

Algorithm

For each row in Seat:

IF id is odd AND id != MAX(id)

→ new_id = id + 1

ELSE IF id is even

→ new_id = id - 1

ELSE (last odd seat)

→ new_id = id (unchanged)

ORDER BY new_id ascending

Time & Space Complexity

	Complexity
TC	$O(N \log N)$ — sorting after computing new ids
SC	$O(N)$ — storing computed ids for each row

Database

Description | Editorial | Solutions | Submissions

626. Exchange Seats

Medium | Topics | Companies

SQL Schema | Pandas Schema

Table: Seat

Column Name	Type
id	int
student	varchar

id is the primary key (unique value) column for this table.
Each row of this table indicates the name and the ID of a student.
The ID sequence always starts from 1 and increments continuously.

Write a solution to swap the seat id of every two consecutive students.
If the number of students is odd, the id of the last student is not swapped.

Return the result table ordered by **id** in ascending order.

The result format is in the following example.

Example 1:

Input:
Seat table:

id	student
1	Abbot
2	Doris
3	Emerson
4	Green

1.9K | 159 | 19 Online

Accepted

All Submissions

Accepted 14 / 14 testcases passed

Sahil Khan submitted at Jun 29, 2026 10:10

Runtime

339 ms Beats 91.64%

Code

MySQL

```
1 # Write your MySQL query statement below
2 SELECT
3     CASE
4         WHEN id%2=1 AND id!=(SELECT MAX(id) FROM Seat) THEN id+1
5         WHEN id % 2 = 0 THEN id - 1
6         ELSE id
7     END AS id,
8     student
9 FROM Seat
10 ORDER BY id
```

View more

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Write your notes here

Code

MySQL

```
1 # Write your MySQL query statement below
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6         ELSE id
7     END AS id,
8     student
9 FROM Seat
10 ORDER BY id
```

Saved

Ln 10, Col 12

Testcase

Test Result

Accepted Runtime: 78 ms

Case 1

Input

Seat =

id	student
1	Abbot
2	Doris
3	Emerson